How to read Statsmodel Regression Summary?

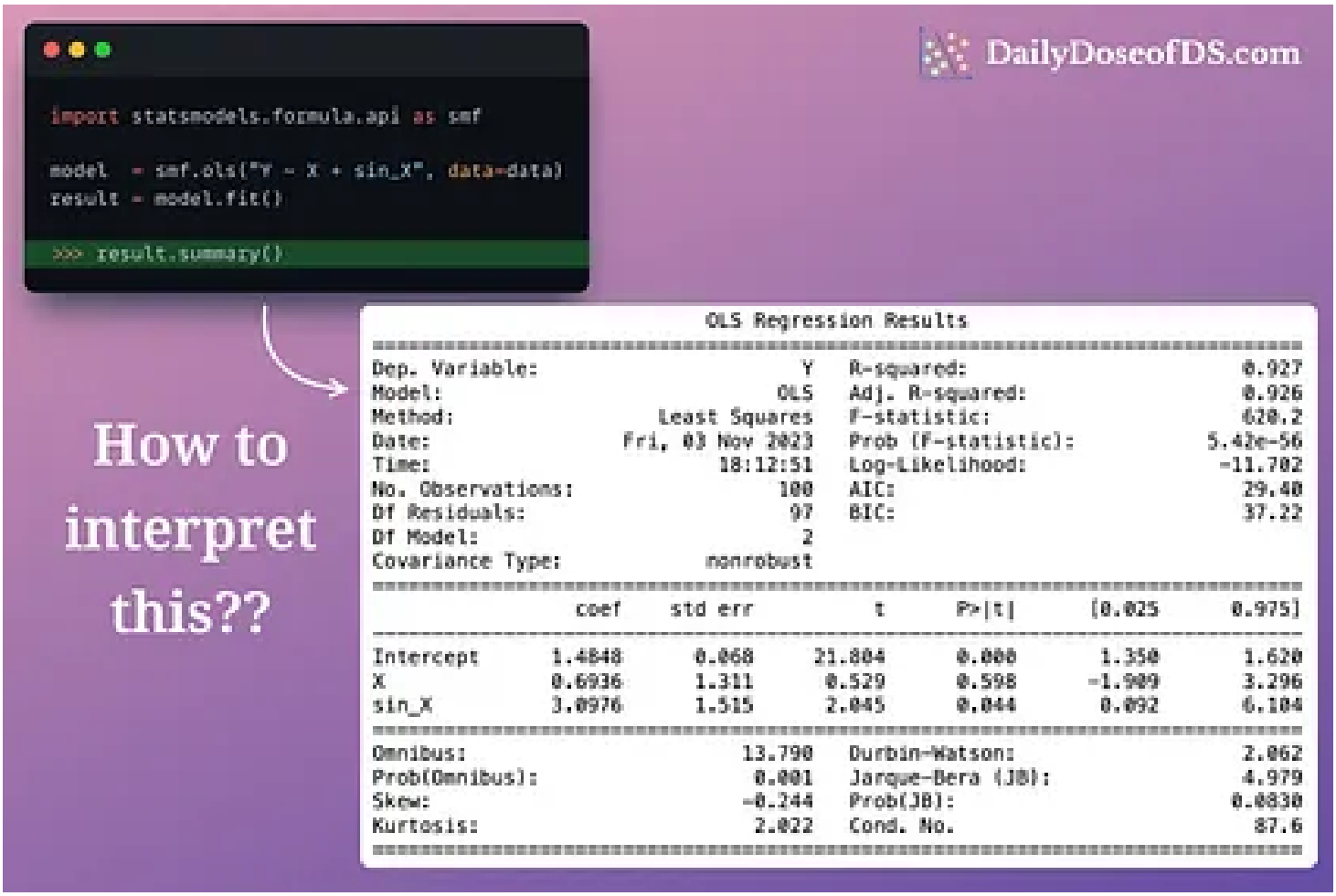
A simplified guide.

AVI CHAWLA

AUG 08, 2024

Daily Dose of Data Science Free Book **/** ***Deep Dives***

Statsmodel provides one of the most comprehensive summaries for regression analysis.

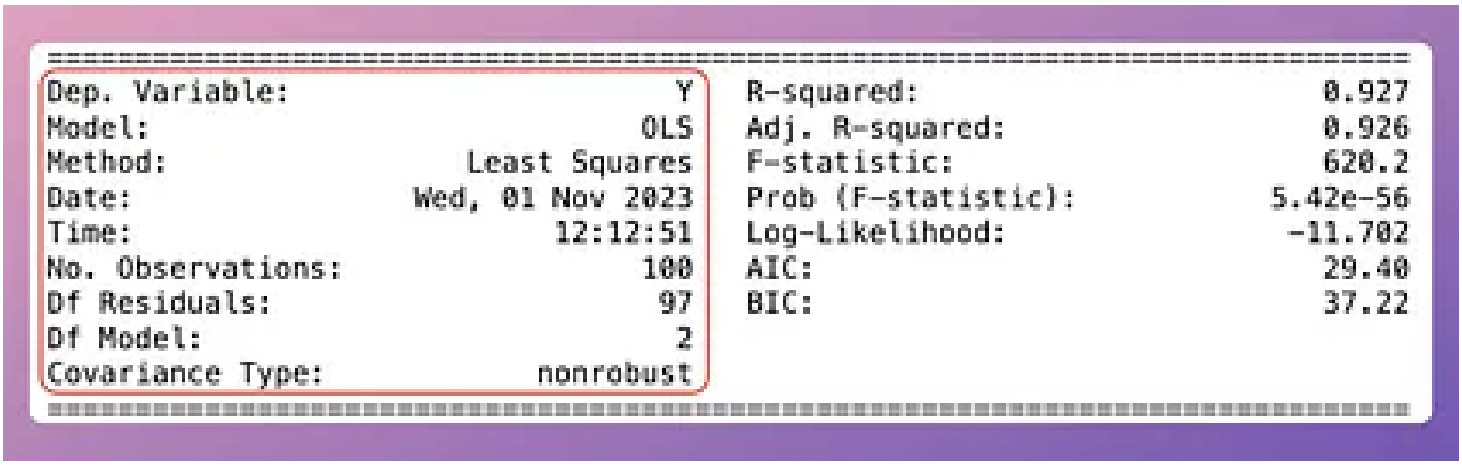


Yet, I have seen so many people struggling to interpret the critical model details mentioned in this report.

Today, let me help you understand the entire summary support provided by statsmodel and why it is so important.

Let's begin!

The first column of the first section lists the model's settings (or config). **This part has nothing to do with the model's performance.**



Dep. Variable:	Y	R-squared:	0.927
Model:	OLS	Adj. R-squared:	0.926
Method:	Least Squares	F-statistic:	620.2
Date:	Wed, 01 Nov 2023	Prob (F-statistic):	5.42e-56
Time:	12:12:51	Log-Likelihood:	-11.702
No. Observations:	100	AIC:	29.40
Df Residuals:	97	BIC:	37.22
Df Model:	2		
Covariance Type:	nonrobust		

- **Dependent variable**: The variable we are predicting.
- **Model and Method**: We are using OLS to fit a linear model.
- **Date and time**: You know it.
- **No. of observations**: The dataset's size.
- **Df residuals**: The degrees of freedom associated with the residuals. It is essentially the number of data points minus the number of parameters estimated in the model (including intercept term).
- **Df Model**: This represents the degrees of freedom associated with the model. It is the number of predictors, 2 in our case — `X` and `sin_X`.

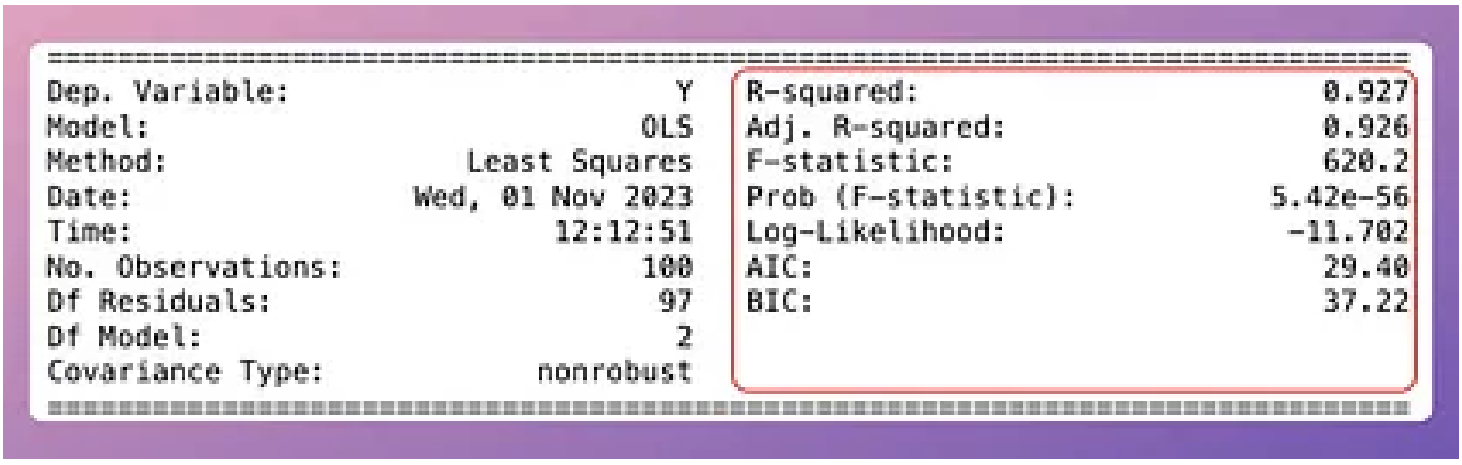
> If your data has categorical features, statsmodel will one-hot encode them. But in that process, it will drop one of the one-hot encoded features.



This is done to avoid the dummy variable trap, which we discussed in this newsletter here.

- **Covariance type**: This is related to the assumptions about the distribution of the residual.
  - In linear regression, we assume that the residuals have a constant variance (homoscedasticity). Why? We discussed it here.
  - We use “nonrobust” to train a model under that assumption.

In the second column, statsmodel provides the overall performance-related details:



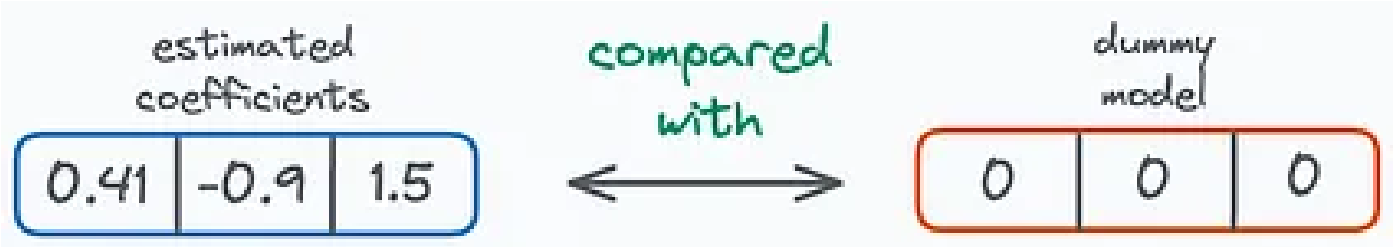
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- **R-squared**: The fraction of original data variability captured by the model.

$$R^2 = \frac{\text{Variability captured by the Model}}{\text{Variability in the original dataset}}$$

- For instance, in this case, 0.927 means that the current model captures 92.7% of the original variability in the **training data**.
- **Statsmodel reports R2 on the input data, so you must not overly optimize for it. If you do, it will lead to overfitting.**

- **Adj. R-squared**:
  - It is somewhat similar to R-squared, but it also accounts for the number of predictors (features) in the model.
  - The problem is that R-squared always increases as we add more features.
  - So even if we add totally irrelevant features, R-squared will never decrease. Adj. R-squared penalizes this misleading behavior of R-squared. More details here.
- **F-statistic and Prob (F-statistic):**
  - These assess the overall **significance** of a regression model.
  - They compare the estimated coefficients by OLS with a model whose all coefficients (except for the intercept) are zero.

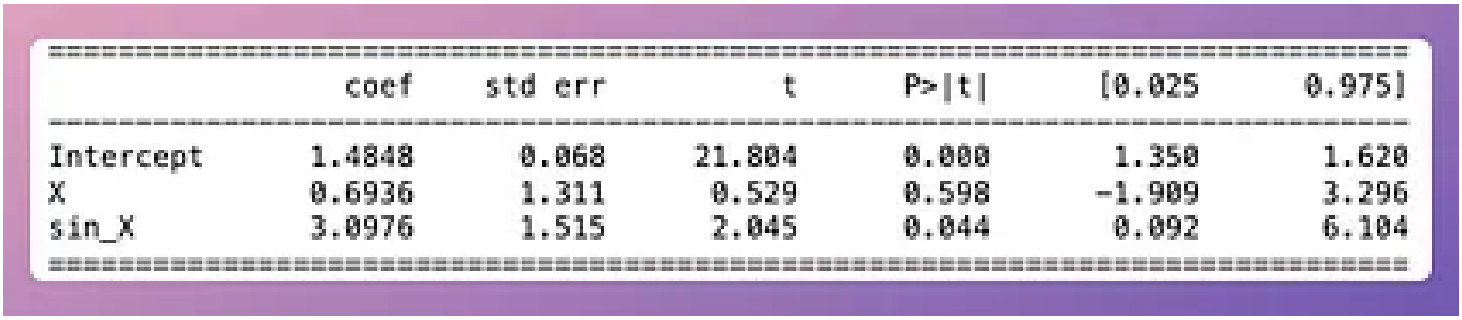


  - **F-statistic** tests whether the independent variables collectively have any effect on the dependent variable or not.
  - **Prob (F-statistic)** is the associated p-value with the **F-statistic.**
    - A small p-value (typically less than 0.05) indicates that the model **as a whole** is statistically significant.
    - This means that at least one independent variable has a significant effect on the dependent variable.
- **Log-Likelihood:**

- This tells us the log-likelihood that the given data was generated by the estimated model.
- The higher the value, the more likely the data was generated by this model.
- We have discussed it multiple times here, here and here.

- **AIC and BIC**:
  - Like adjusted R-squared, these are performance metrics to determine goodness of fit while penalizing complexity.
  - Lower AIC and BIC values indicate a better fit. We discussed them here.

The second section provides details related to the features:

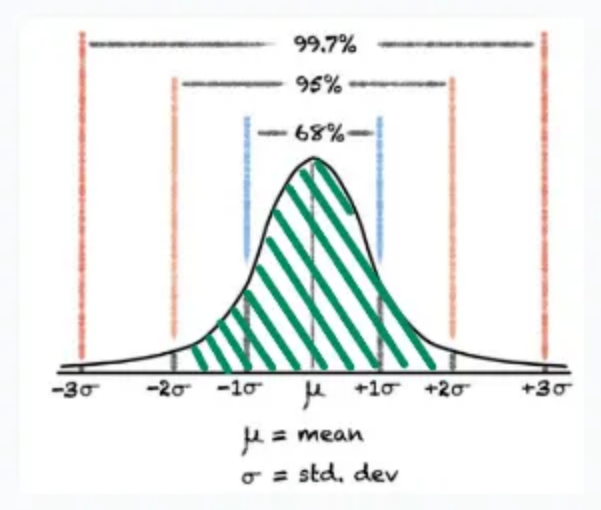


	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	1.4848	0.068	21.804	0.000	1.350	1.620
X	0.6936	1.311	0.529	0.598	-1.909	3.296
sin_X	3.0976	1.515	2.045	0.044	0.092	6.104

- **coef:** The estimated coefficient for a feature.
- **t and P>|t|:**
  - Earlier, we used **F-statistic** to determine the statistical significance of the model **as a whole**.
  - **t-statistic** is more granular on that front as it determines the significance of **every individual feature**.
  - **P>|t|** is the associated p-value with the **t-statistic.**
  - A small p-value (typically less than 0.05) indicates that the feature is statistically significant.

- For instance, the feature “X” has a p-value of ~0.6. This suggests that there is a 60% chance that the feature “X” has no effect on “Y”.

- **[0.025, 0.975]** and **std err:**
  - See, the coefficients we have obtained from the model are just estimates. They may not be **absolute true coefficients** of the process that generated the data.
  - Thus, the estimated parameters are subject to uncertainty, aren’t they?
  - Note: The width of the interval **[0.025, 0.975]** is **0.95** → or 95%. This constitutes the area between 2 standard deviations from the mean in a normal distribution**.**



  - A 95% confidence interval provides a range of values within which you can be **95% confident** that the true value of the parameter lies.

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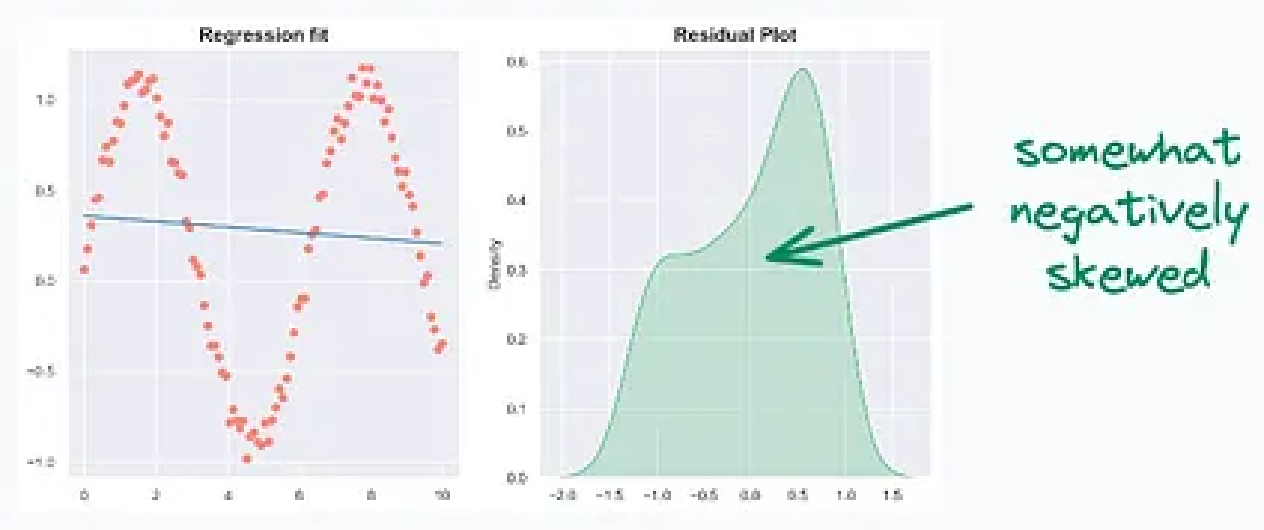
  - For instance, the interval for `sin_X` is `(0.092, 6.104)`. So although the estimated coefficient is `3.09`, we can be 95% confident that the true coefficient lies in the range `(0.092, 6.104)`.
  - What is a confidence interval? We covered it here.

Details in the last section of the report test the assumptions of linear regression.



Omnibus:	13.790	Durbin-Watson:	2.062
Prob(Omnibus):	0.001	Jarque-Bera (JB):	4.979
Skew:	-0.244	Prob(JB):	0.0830
Kurtosis:	2.022	Cond. No.	87.6

- **Omnibus and Prob(Omnibus)**:
  - They test the normality of the residuals.
  - **Omnibus** value of zero means residuals are perfectly normal.
  - **Prob(Omnibus)** is the corresponding p-value.
    - In this case, **Prob(Omnibus)** is 0.001. This means there is a 0.1% chance that the residuals are normally distributed.
- **Skew and Kurtosis:**
  - They also provide information about the distribution of the residuals.
  - Skewness measures the asymmetry of the distribution of residuals.
    - Zero skewness means perfect symmetry.
    - Positive skewness indicates a distribution with a long right tail. This indicates a concentration of residuals on lower values. Good to check for outliers in this case.
    - Negative skewness indicates a distribution with a long left tail. This is mostly indicative of poor features. For instance, consider fitting a sin curve with a linear feature (X). Most residuals will be high, resulting in negative skewness.



- **Durbin-Watson:**
  - This measures autocorrelation between residuals.
  - Autocorrelation occurs when the residuals are correlated, indicating that the error terms are not independent.
  - But linear regression assumes that residuals are not correlated.
  - The Durbin-Watson statistic ranges between `0` and `4`.
    - A value close to 2 indicates no autocorrelation.
    - Values closer to 0 indicate positive autocorrelation.
    - Values closer to 4 indicate negative autocorrelation.
- **Jarque-Bera (JB) and Prob(JB):**
  - They solve the same purpose as **Omnibus and Prob(Omnibus)** — measuring the normality of residuals.
- **Condition Number**:
  - This tests multicollinearity.
  - Multicollinearity occurs when two features are correlated, or two or more features determine the value of another feature.
  - A standalone value for **Condition Number** can be difficult to interpret so here's how I use it:
    - Add features one by one to the regression model and notice any spikes in the **Condition Number.**